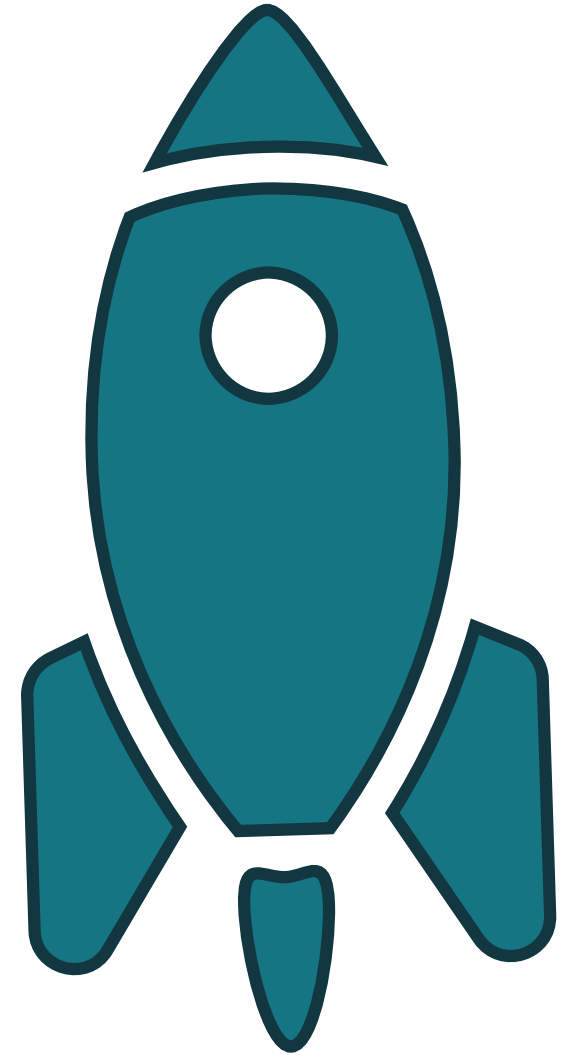


PHS 2000 Math Review

Ashlynn Wimer

August 2026



Your knowledge is
about to take off

Happy noonish!

Welcome to math review!

I'll be covering some key concepts that will be helpful for building intuition for the methods we learn about in PHS2000.

Reminders

- Everyone comes into PHS with different backgrounds (which is what makes PHS so special!)
- You don't need to memorize everything in this session
- These slides will be available to you as a reference

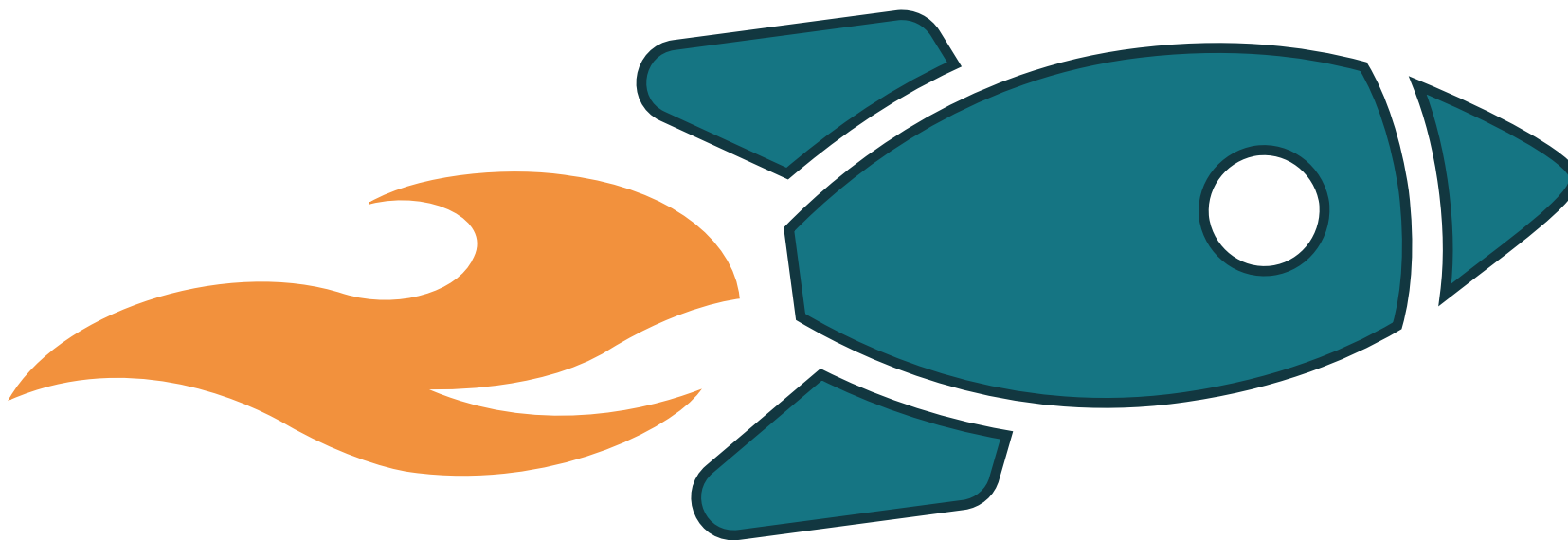


Outline

Today's objective is to define some common functions and to apply them to concepts we will encounter in PHS2000A :

- Functions
 - Exponential & Logarithmic
 - Expit & Logit
- Probability & odds
- Calculus lite TM
- Applied example

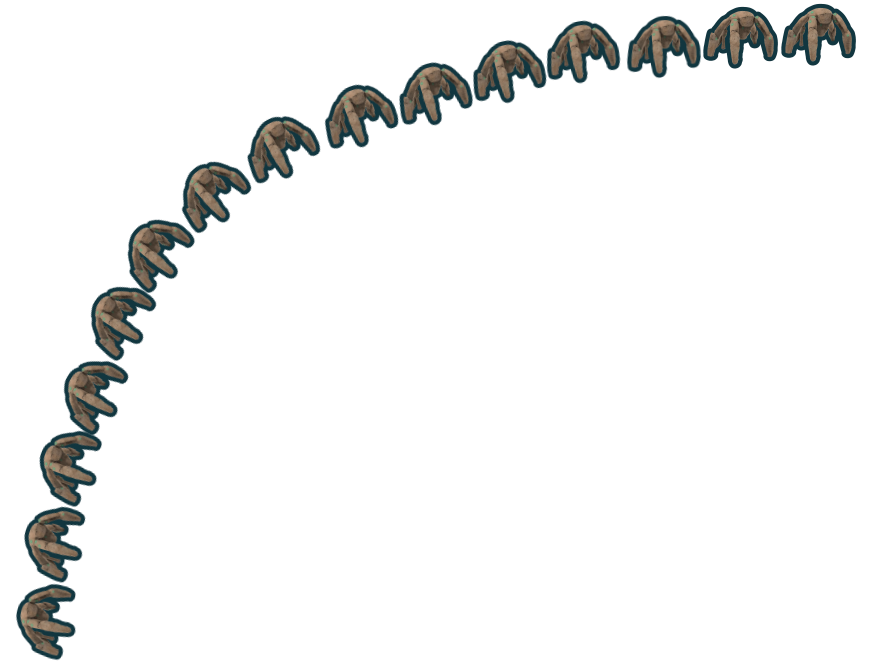
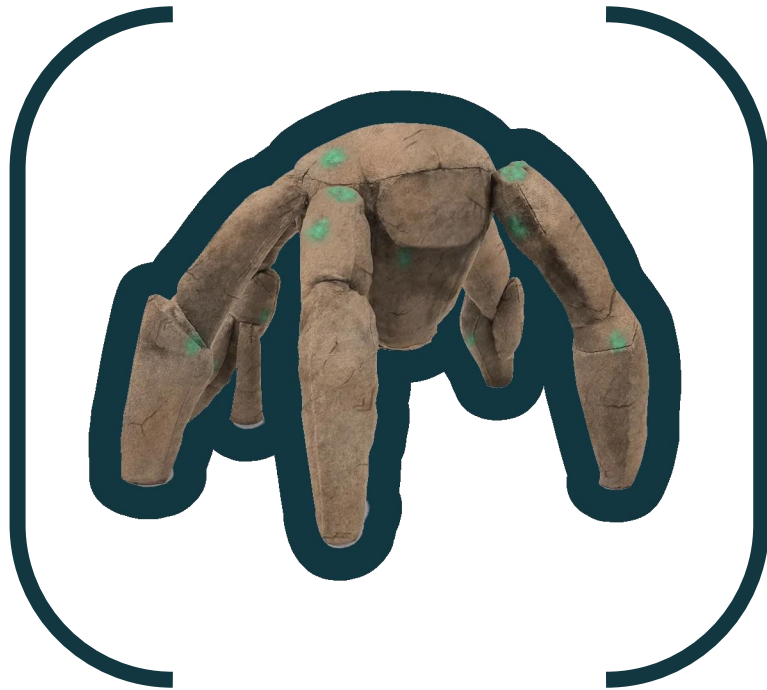
Liftoff!!



Please stop me whenever to ask questions!
(I have a habit of rambling and talking fast. So please, ask me questions!).

Functions, probability, and odds

log

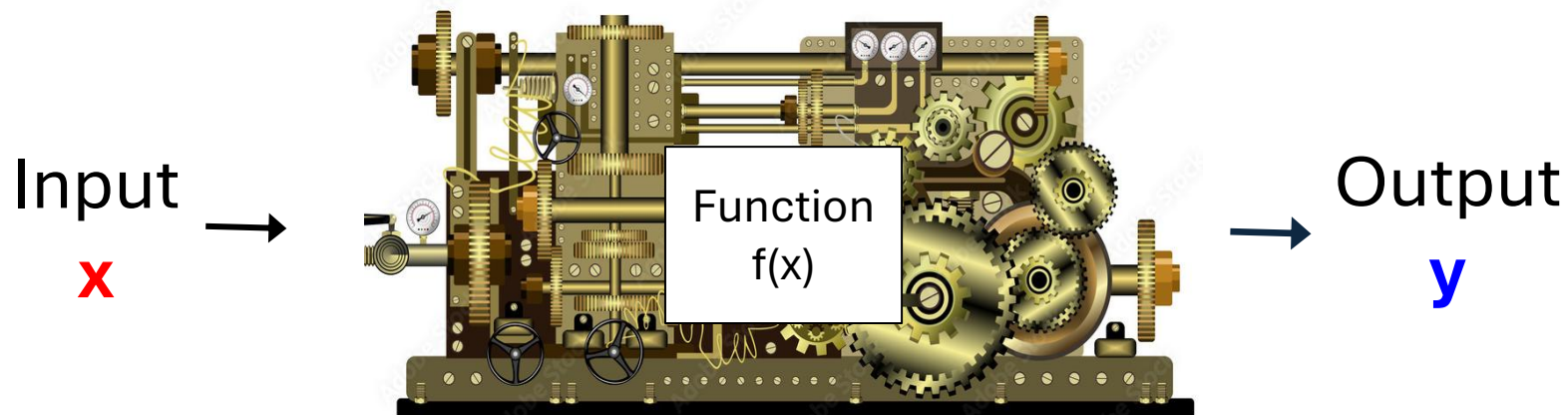


Functions

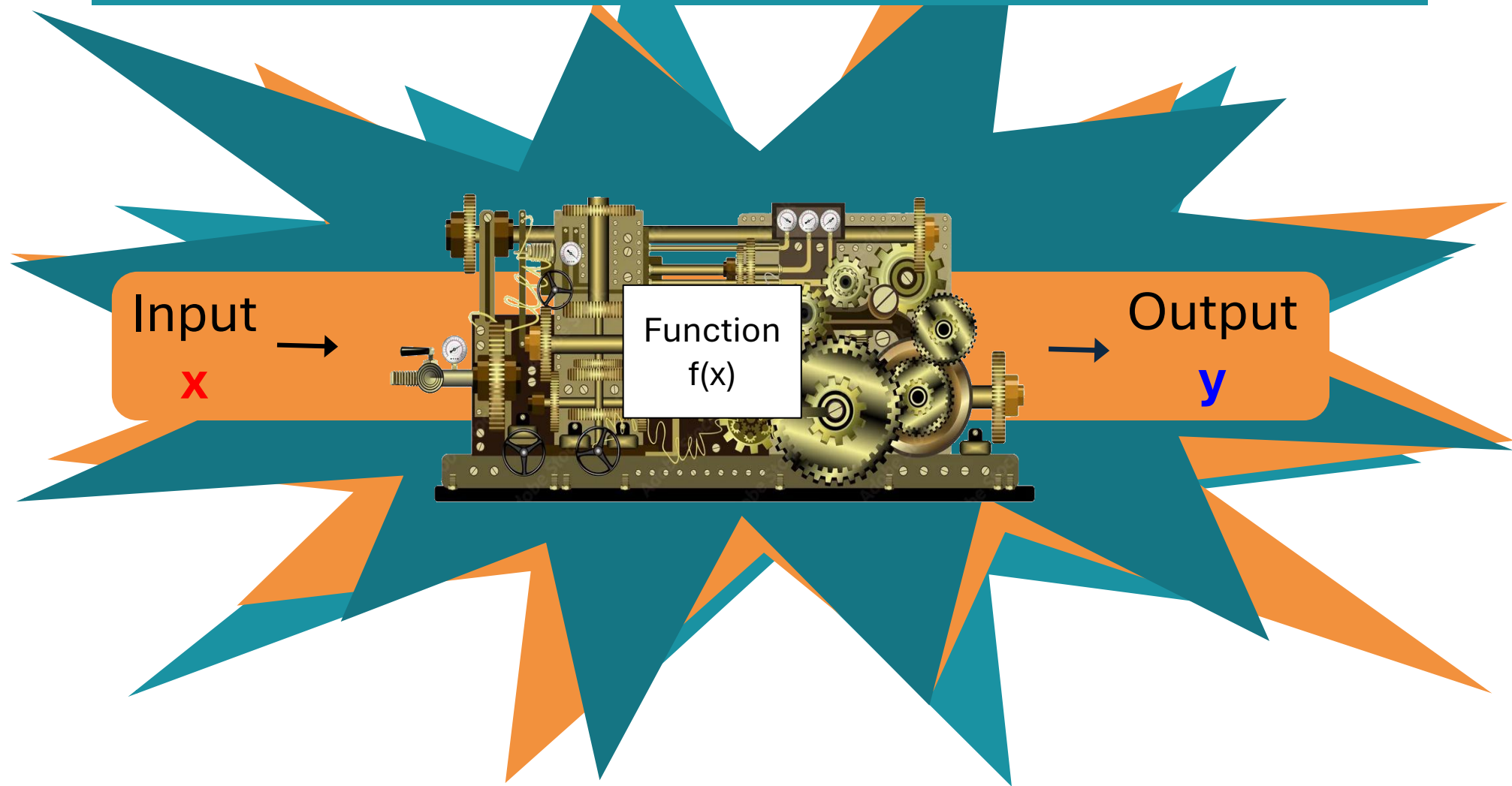
A procedure that takes an input, **x**, and gives you an output, **y**.

Each **input** “**x**” has only a single **output** “**y**”, i.e., every **x** uniquely produces a particular **y**.

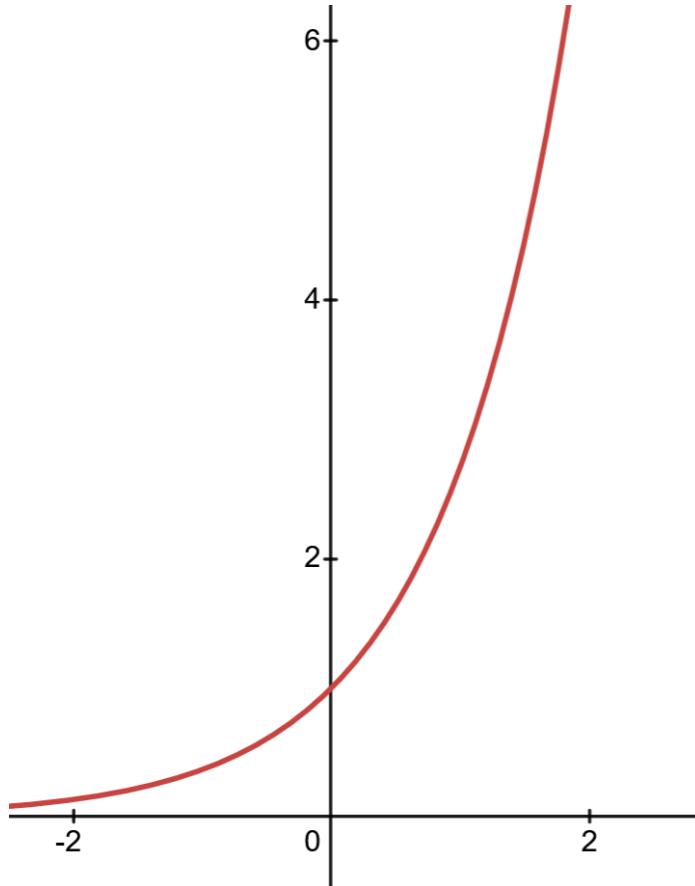
We call the set of all possible input values **domain** of a function, and the set of all possible outputs the **range** of the function.



Let's Look At Really Cool Functions!



Natural Exponential Function!



The **natural exponential** function:

$$y = \exp(x) = e^x$$

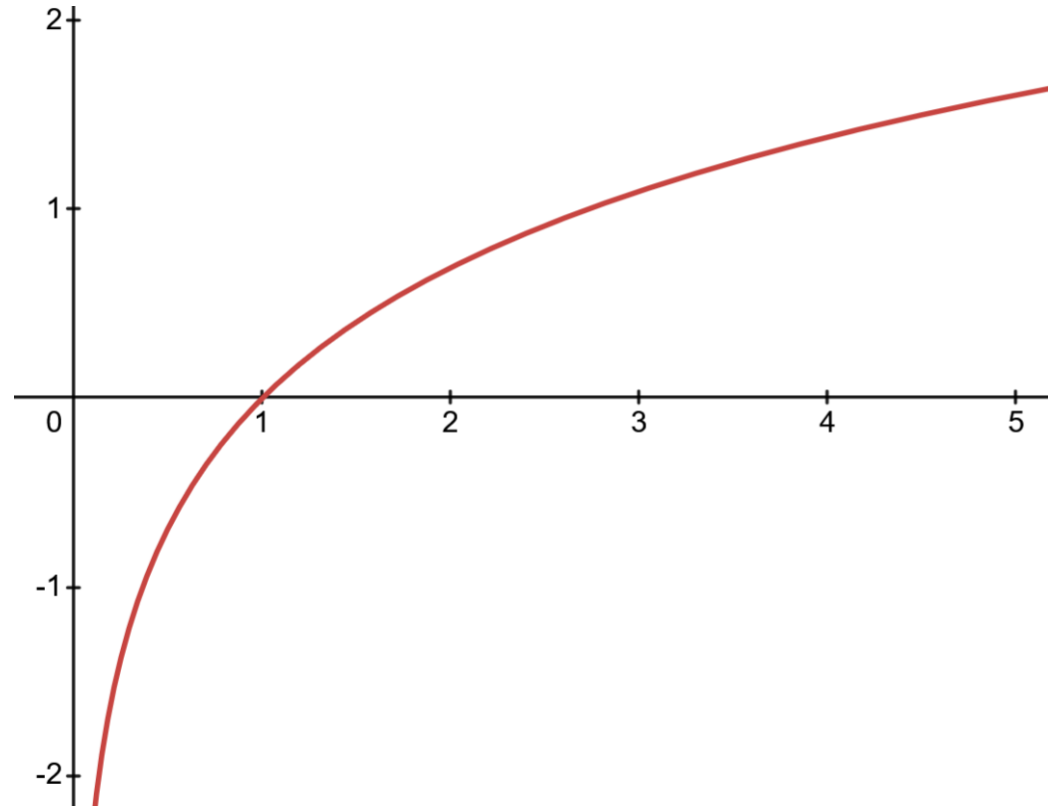
- **e** stands for **Euler's constant** $e \sim 2.718$
- The function is always positive, >0 .
- Domain: $(-\infty, +\infty)$
- Range: $(0, +\infty)$
- Note: $\exp(0) = 1$

Natural Log Functions

The **natural log** function:

$$y = \log(x) = \ln(x)$$

- Domain: $(0, +\infty)$
- Range: $(-\infty, +\infty)$
- $\log(x)$ refers to the natural logarithm (i.e., $\ln$) with base e in this class

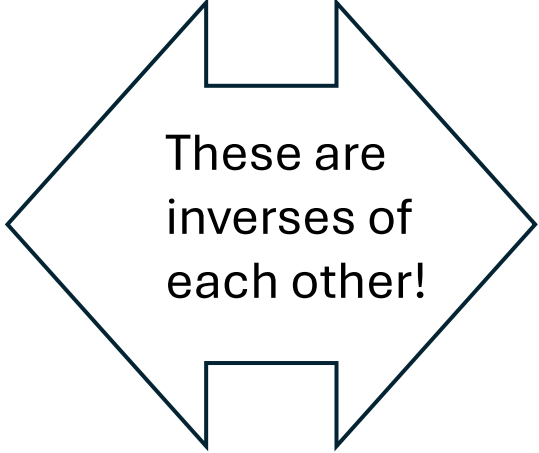


Natural log and natural exponential functions

The **natural log** function:

$$y = \log(x) = \ln(x)$$

- Domain: $[0, +\infty]$
- Range: $[-\infty, +\infty]$
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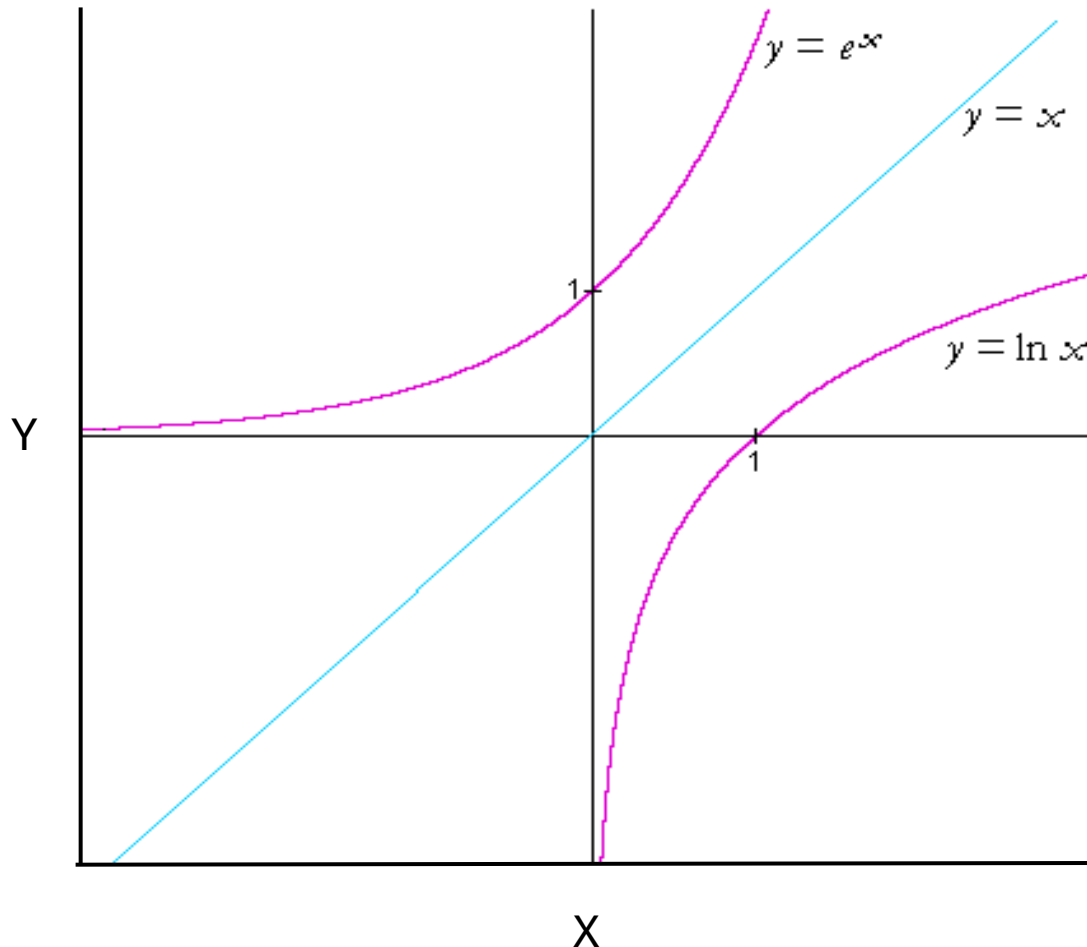
These are
inverses of
each other!

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Natural log and natural exponential functions



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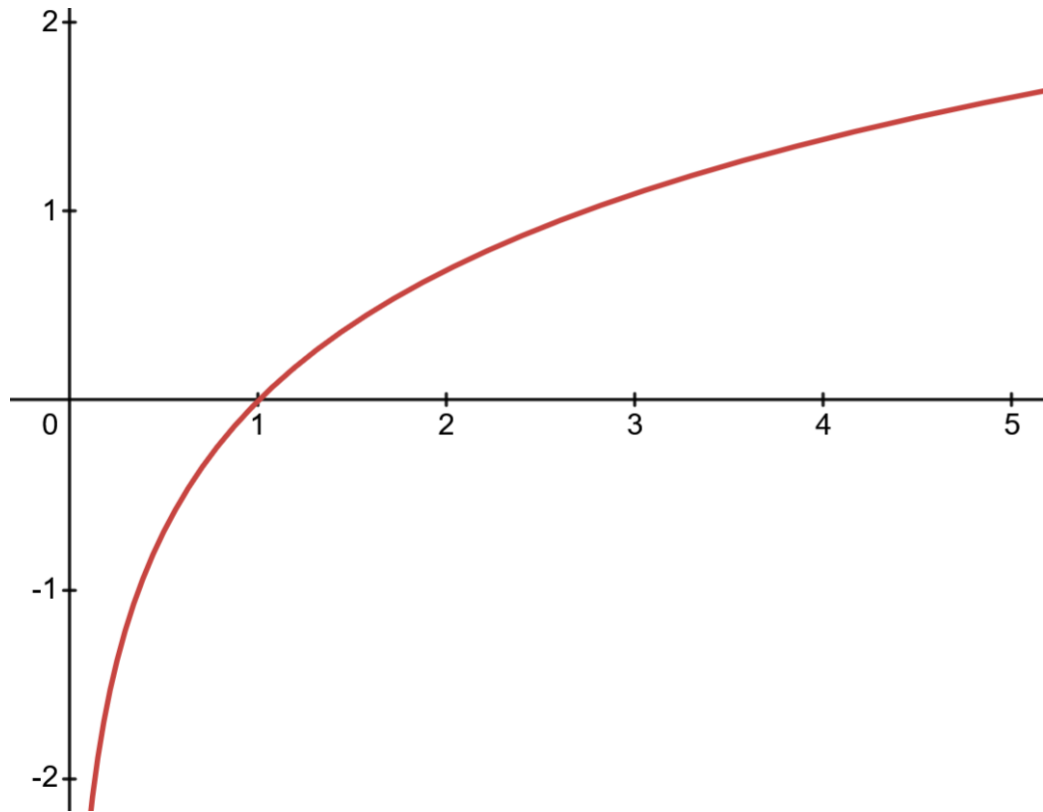
$$y = \exp(x) = e^x$$

- Domain: $[-\infty, +\infty]$
- Range: $[0, +\infty]$

Log & exponent rules

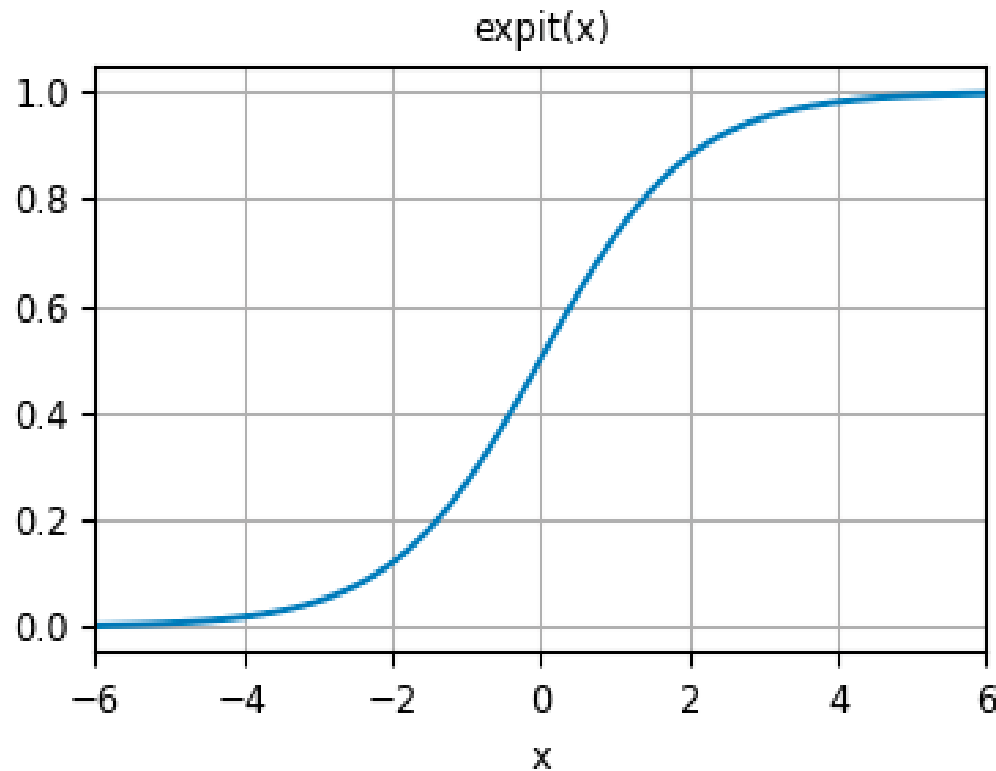
Laws of Exponents	Laws of Logarithms
$\exp(a + b) = \exp(a) \times \exp(b)$ $\rightarrow e^{a+b} = e^a \times e^b$	$\log(a) + \log(b) = \log(a \times b)$
$\exp(a-b) = \exp(a) / \exp(b)$ $\rightarrow e^{a-b} = e^a / e^b$	$\log(a) - \log(b) = \log\left(\frac{a}{b}\right)$
$\exp(a)^b = \exp(a \times b)$ $\rightarrow e^{a^b} = e^{a \times b}$	$\log(a^b) = b \times \log(a)$

Pop Quiz!



What function is this?
(shout it out!)

Logit and expit functions



The **expit** function:

$$y = \text{expit}(x) = \frac{e^x}{1 + e^x}$$

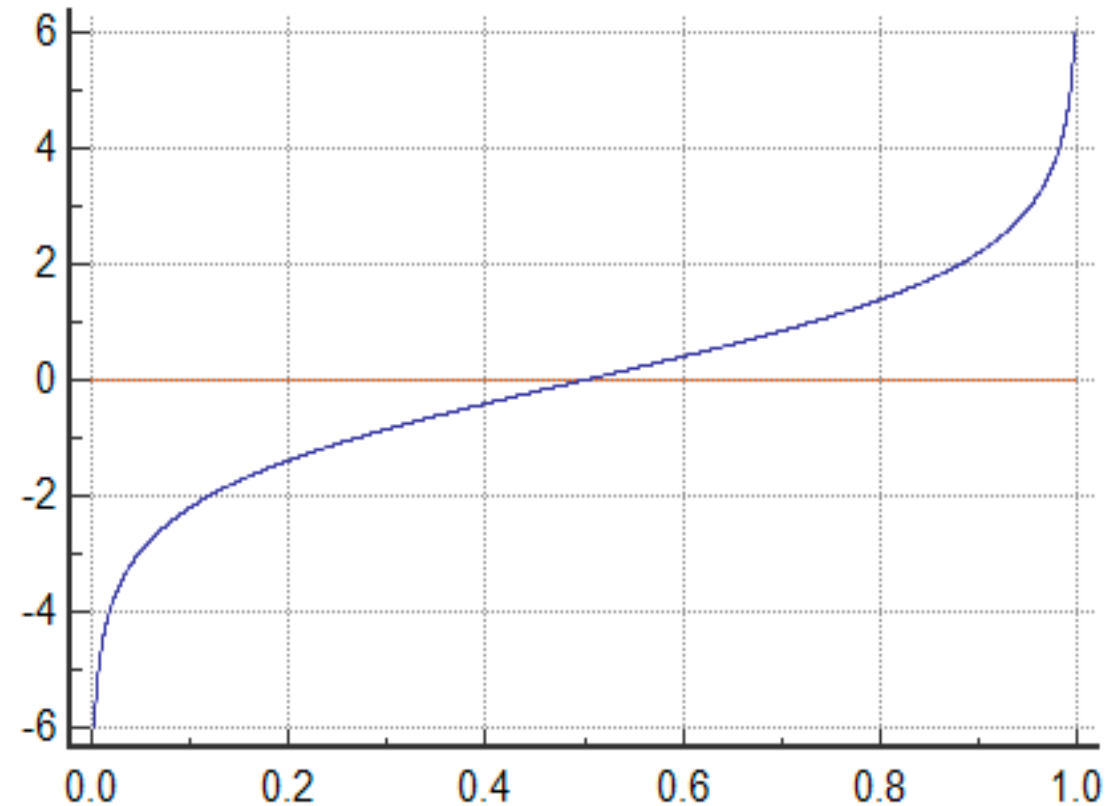
- Domain: $(-\infty, +\infty)$
- Range: $(0, 1)$
- Often used when we want to **transform data to a range that resembles probabilities / proportions** between 0 and 1.

Logit and expit functions

The **logit** function:

$$y = \text{logit}(x) = \log\left(\frac{x}{1-x}\right)$$

- Domain: $(0, 1)$
- Range: $(-\infty, +\infty)$
- Commonly encountered in logistic regression to convert probabilities back to a wider range of real numbers.

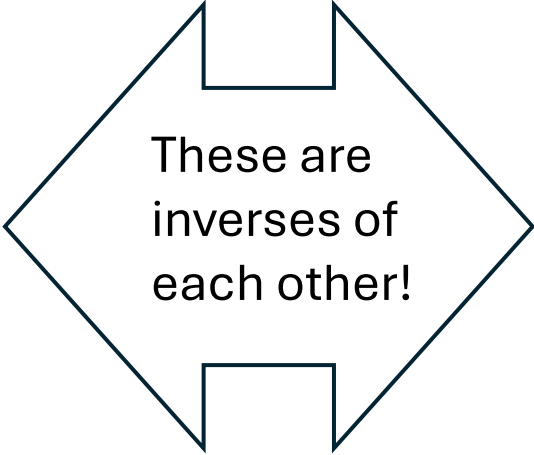


Logit and expit functions

The **logit** function:

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- Domain: $[0, 1]$
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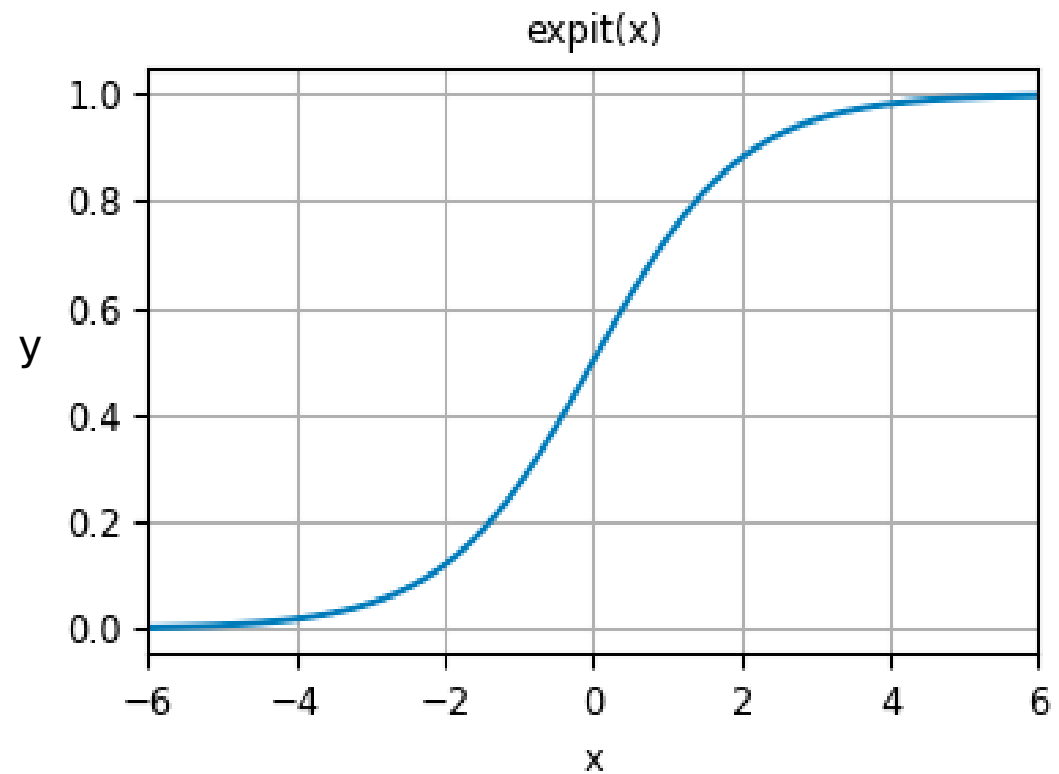
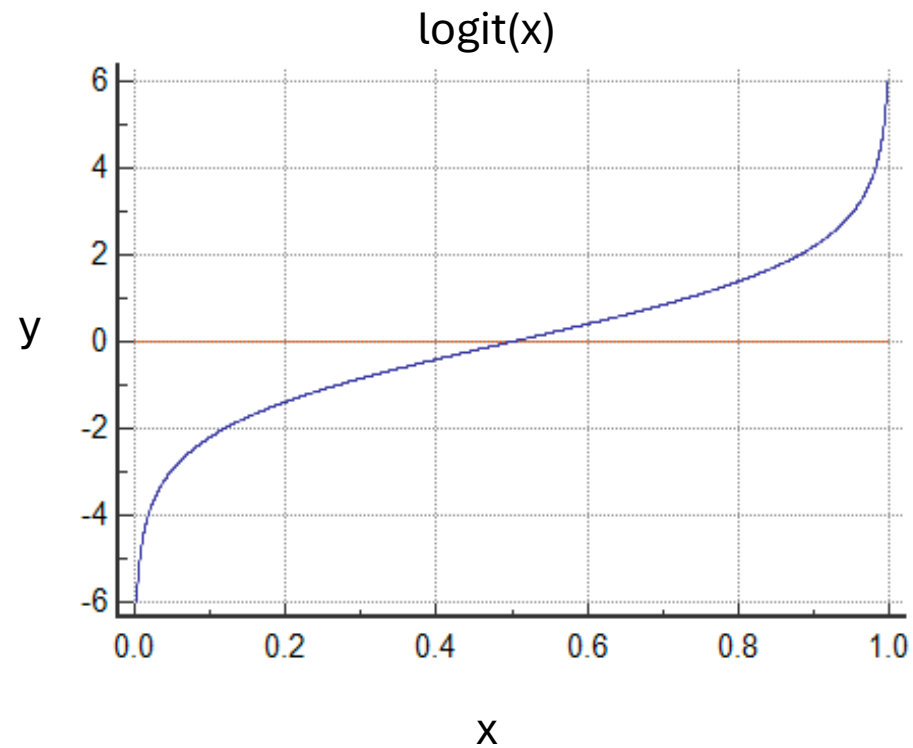
These are
inverses of
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The **expit** function:

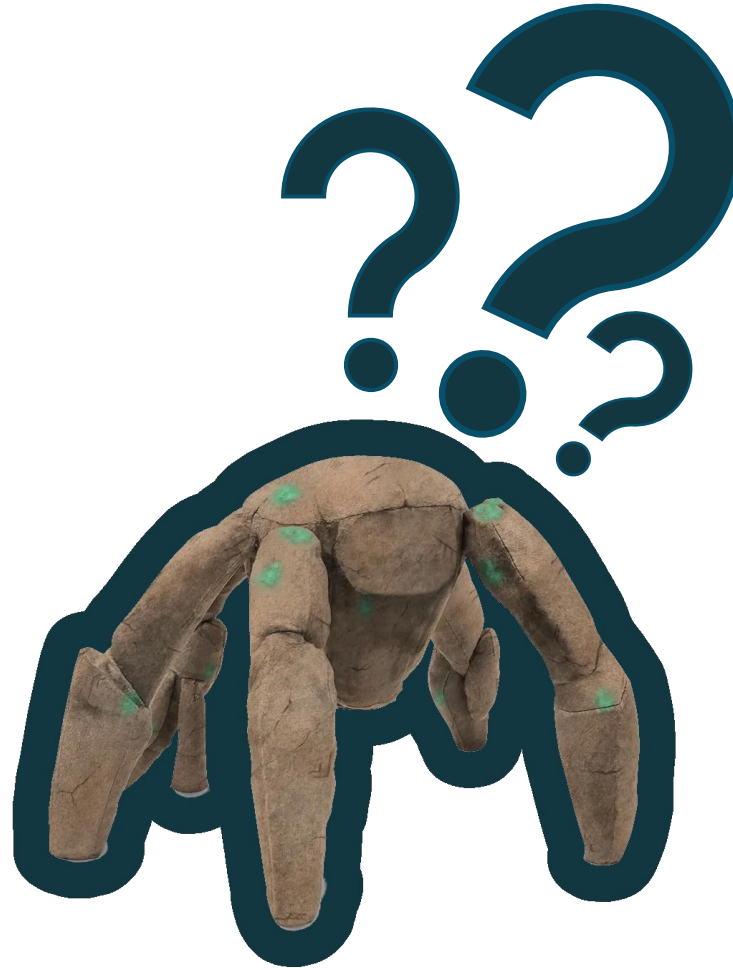
$$y = \text{expit}(x) = \frac{e^x}{1 + e^x}$$

- Often used when we want to **transform data to a range that resembles probabilities / proportions** between 0 and 1.
- Domain: $(-\infty, +\infty)$
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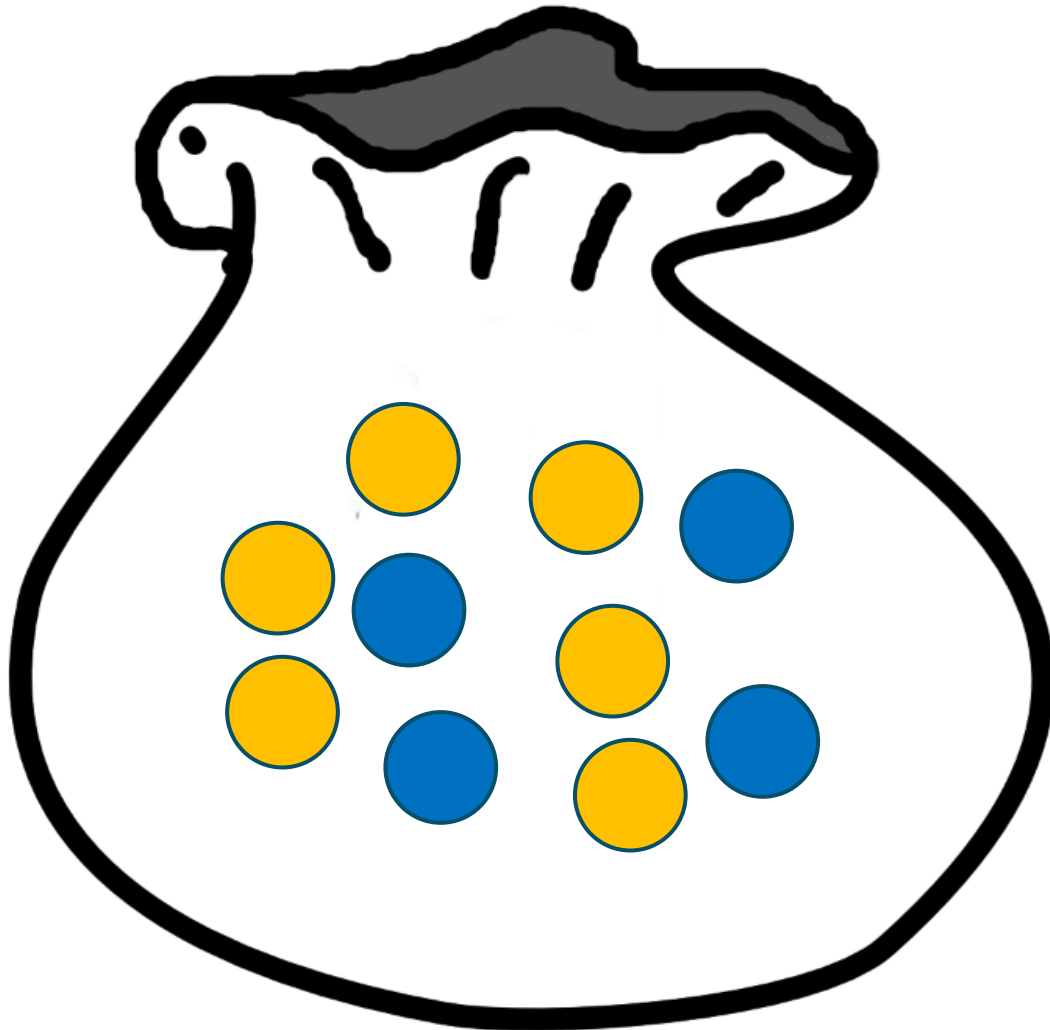
Logit and expit functions



Questions?



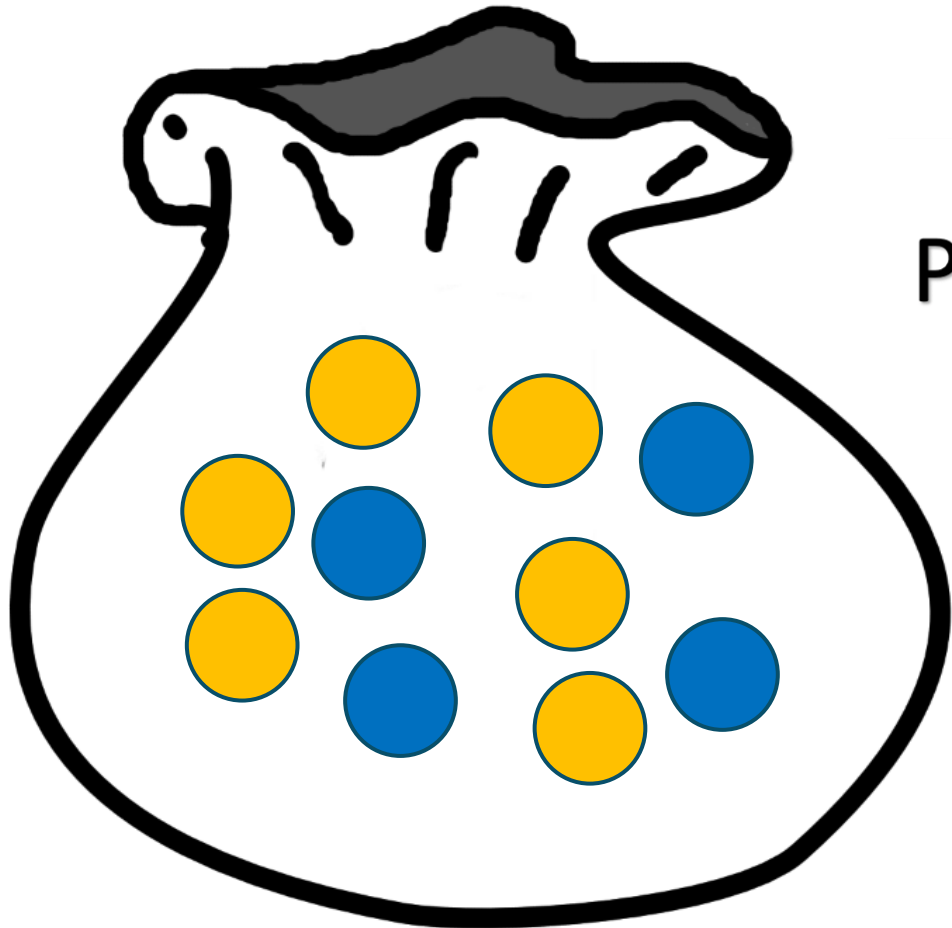
Probability and Odds



Suppose we have a sack of 6 yellow space rocks and 4 blue space rocks.

We randomly draw a space rock from the sack

What is the probability we draw a blue space rock?

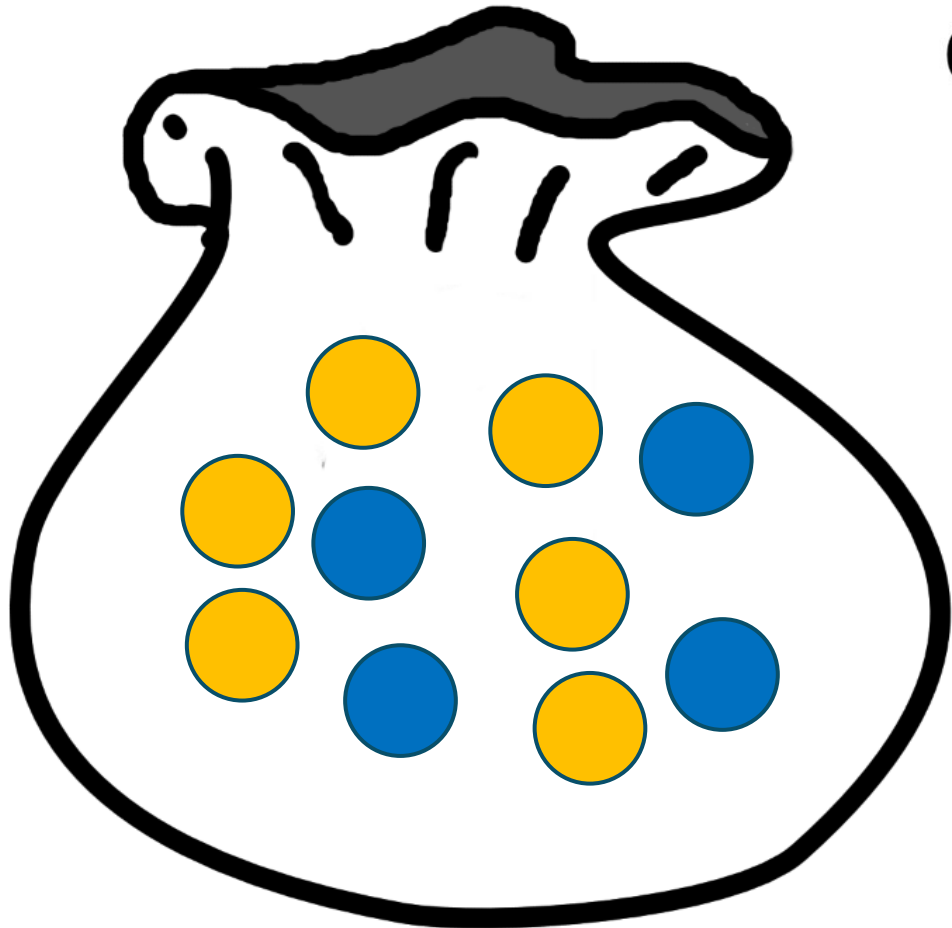


$$\text{Probability} = \frac{\text{4 blue rocks}}{\text{4 blue rocks} + \text{6 yellow rocks}}$$

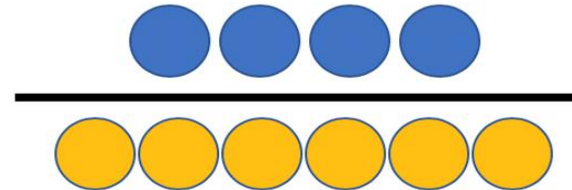
Note that we are always dividing a **part** by the **whole**.

As such, probabilities **must be between 0 and 1**

What is the odds we draw a blue space rock?



Odds =



Note that we are taking the **ratio** of two events (successes versus failures).

These are always **positive**, but **unbounded**, so **odds range from 0 to positive infinity!**

Odds **less than one** means that we **fail more** often than we succeed.

Odds **greater than one** means that we **succeed more** often than we fail!

Probability and odds

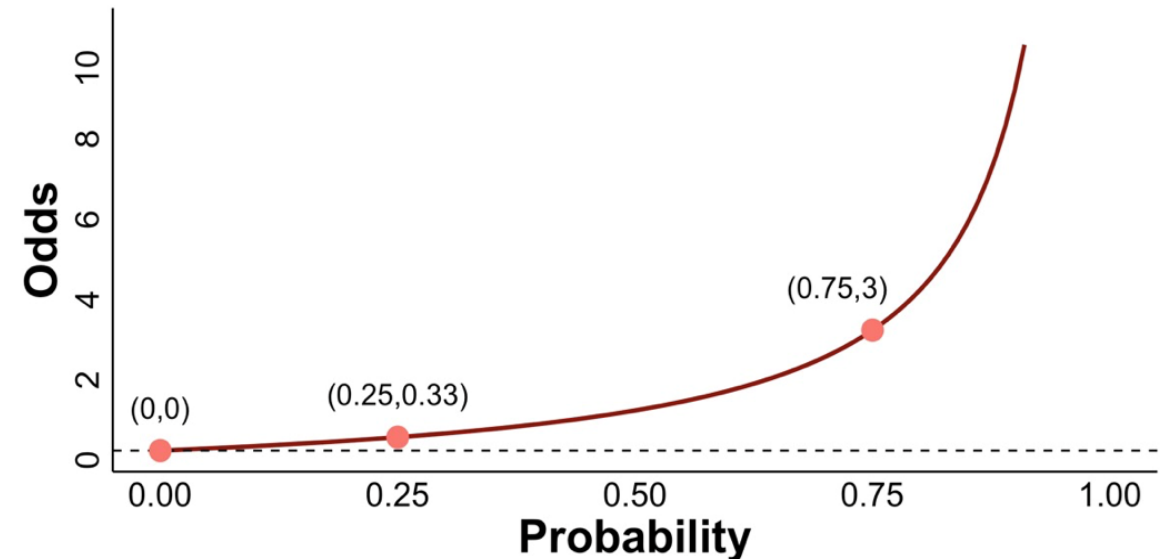
Relationship between probability and odds:

$$\text{Probability} = \frac{\text{odds}}{(1 + \text{odds})}$$

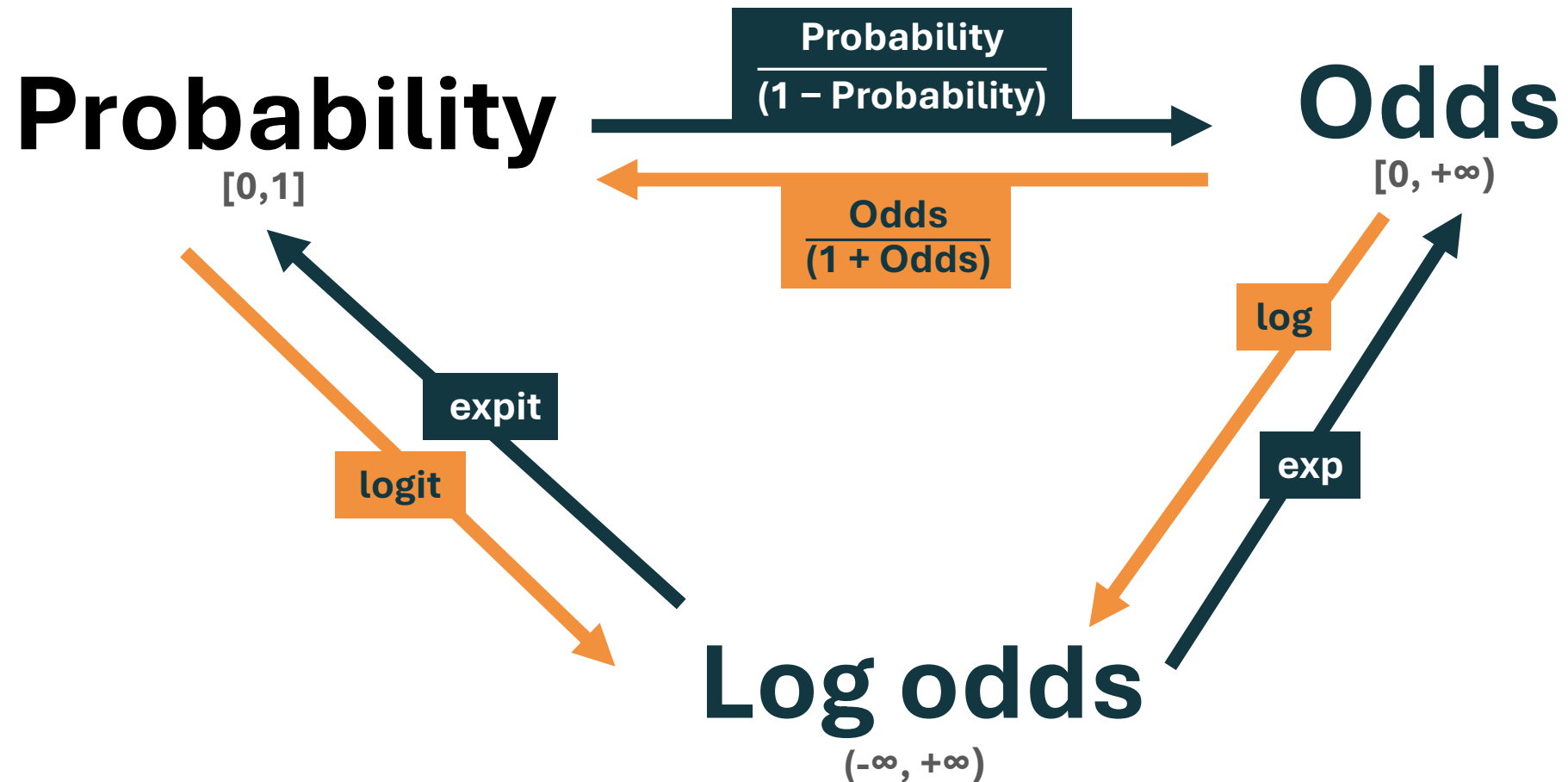
$$\text{Odds} = \frac{\text{Probability}}{(1 - \text{probability})}$$

Looks a bit
like the logit
and expit
functions!

As probabilities get larger, the difference between probability and odds **increases**.



Bringing it together:



Bringing it together: logistic regression

If we want to model the *probability* something happens, it's easier to instead use the **logit** function to model the **log odds** it does.

Probability

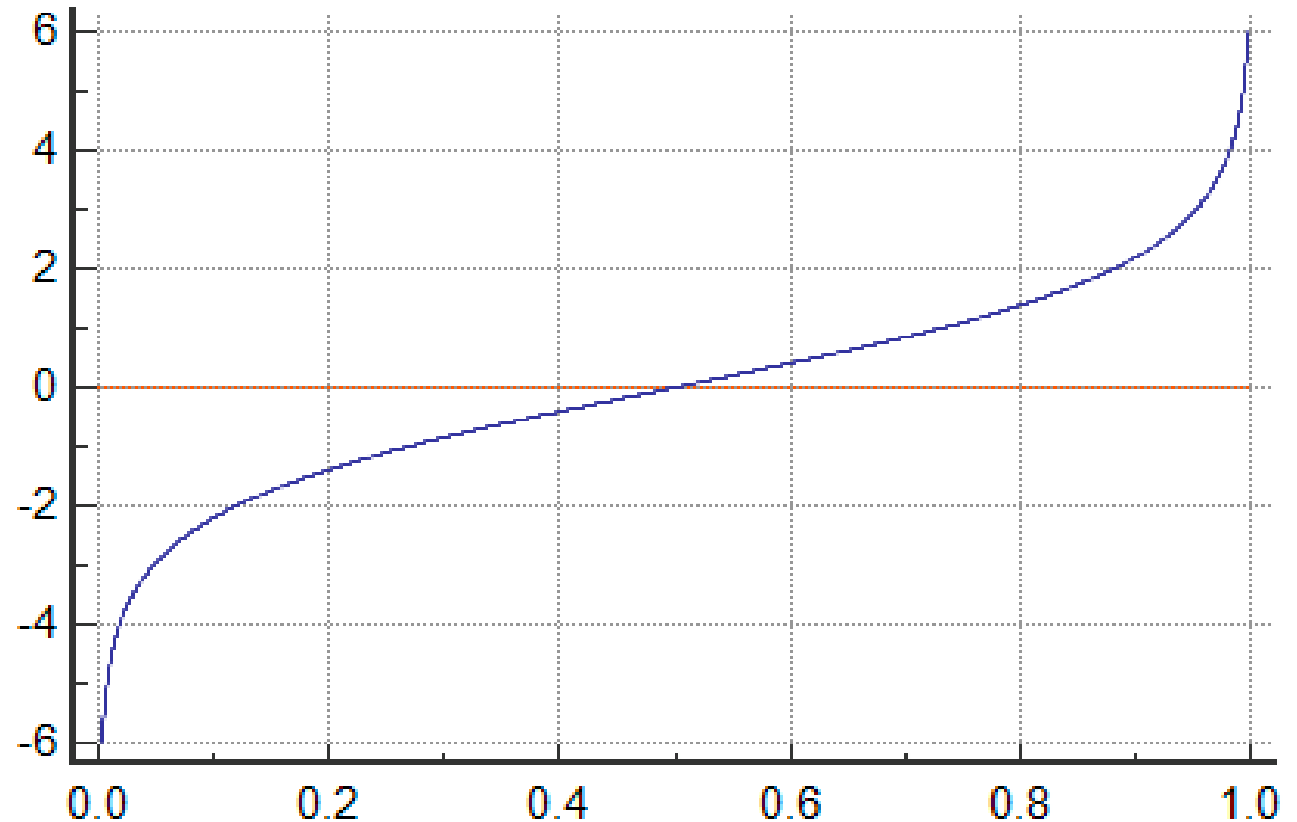
$[0,1]$

logit

expit

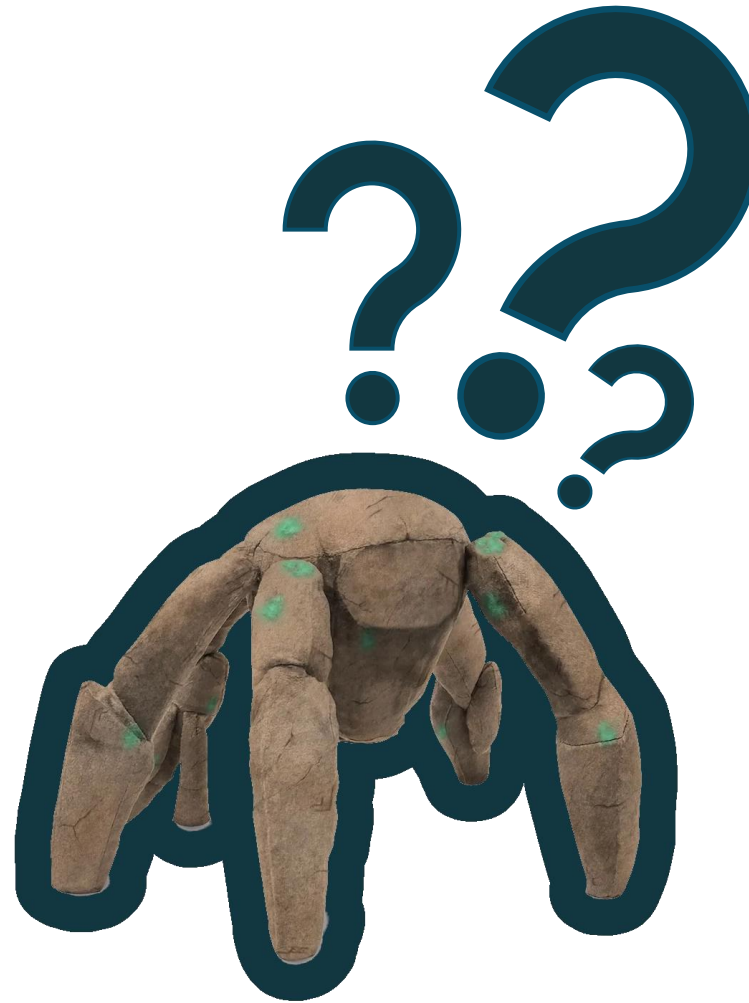
Log odds

$(-\infty, +\infty)$





Questions?



Calculus-lite

Calculus is not **integral** to your understanding, but it certainly helps.



What is calculus?

Calculus is the study of **differentiation** and **integration**, which are rooted in the idea of limits and functions!

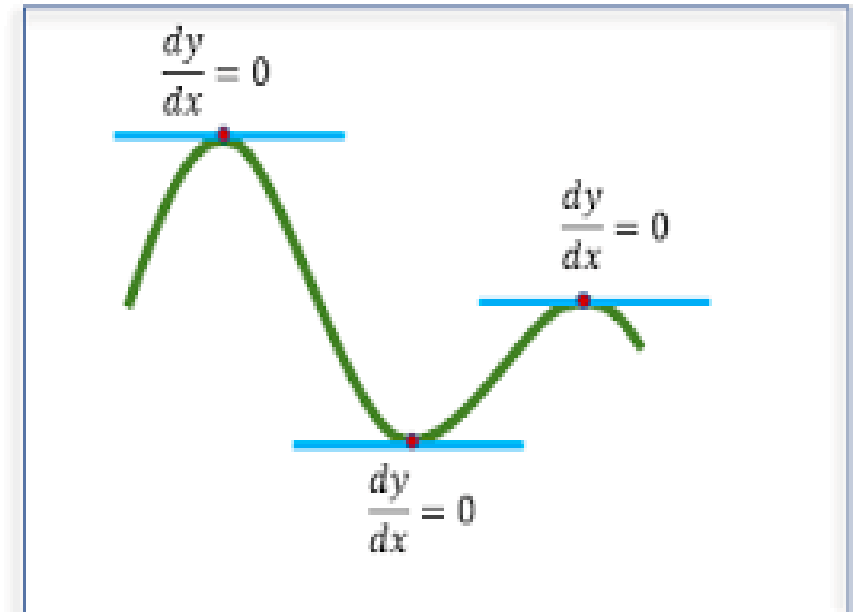
Calculus deals with the properties of derivatives and integrals.

- **Differential calculus** = Find the rate of change of a quantity
- **Integral calculus** = Find the quantity when the rate of change is unknown

Differential calculus

- The process of finding derivatives is called **differentiation**.
- **The derivative of a function** is represented by: $\frac{dy}{dx}$ or $f'(x)$
- Derivatives find the **maxima** and **minima** values of a function.
 - **Maxima and minima are where $f'(x) = 0$**

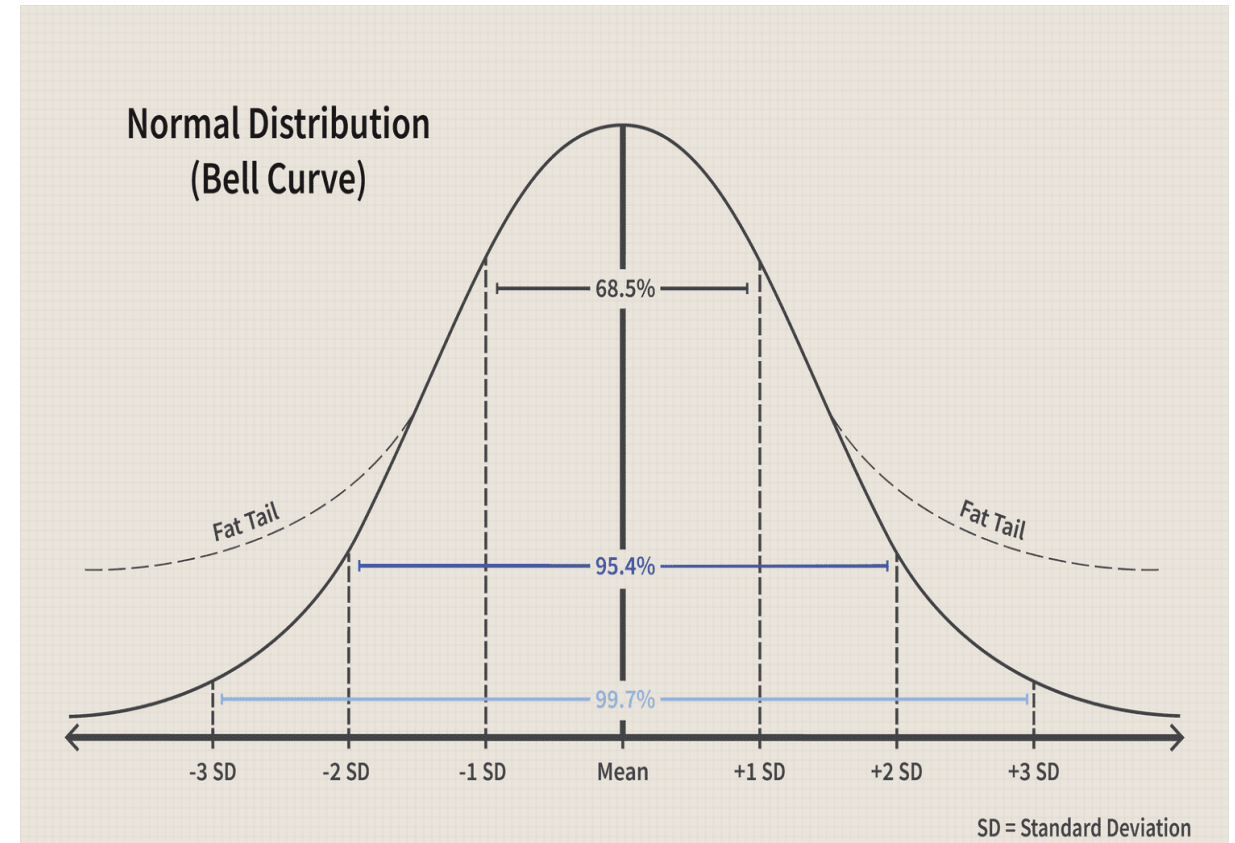
Sneak peak: Calculus is used to find the maximum point of the log-likelihood function when we learn about Maximum Likelihood Estimation (MLE).



Integral calculus

Integration = Reciprocal of differentiation.

- The goal of integral calculus is to find the area under a curve over a defined interval.
- **Sneak peak:** *Integrals (and derivatives) are helpful when dealing with probability density functions (PDF) and cumulative distribution functions (CDF).*



Questions?

Project Hail Mary Creators Answer The 50 Most Searched PHS Calculus-lite Questions



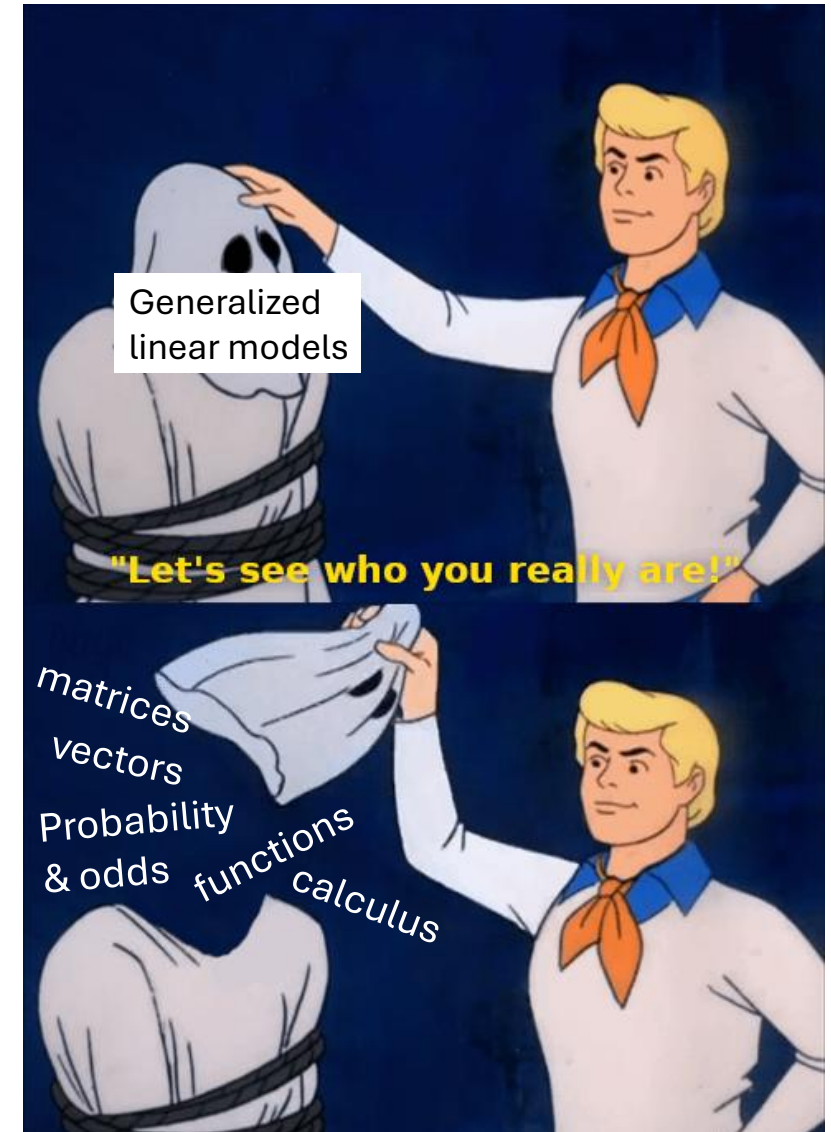


Example

Regression example

Now we're going to do some modeling!

As we go through this example, I'll be asking how different components of the model work – and as a hint, I've already given you all the answers!



Regression example

You and your friend are bored at Kresge.

After conducting some primary data collection among your compatriots, you want to model the **probability** that **someone has read Project Hail Mary** as a function of:

- their **age**,
- whether they have **watched the movie**,
- whether they have **an AO3 account**,
- the number of **pages** they (purportedly) read last year.

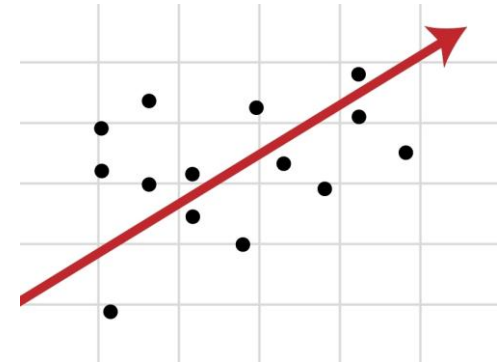
Regression example

We want our model to look something like this:

$$\pi(x) = y = \beta_0 + \beta_1 \text{age} + \beta_2 \text{movie} + \beta_3 (\text{AO3}) + \beta_4 \text{pages}$$

- **π** : probability that someone has read Project Hail Mary
- **x** : vector of variables (*age*, *movie*, *ao3*, *pages*)
- **age**: age in years
- **movie**: whether they have seen *Project Hail Mary* (0: no, 1: yes)
- **AO3**: whether they have used AO3 this month (0: no, 1: yes)
- **pages**: the number of pages they have read in the past year

What issues do we spot with this model specification?

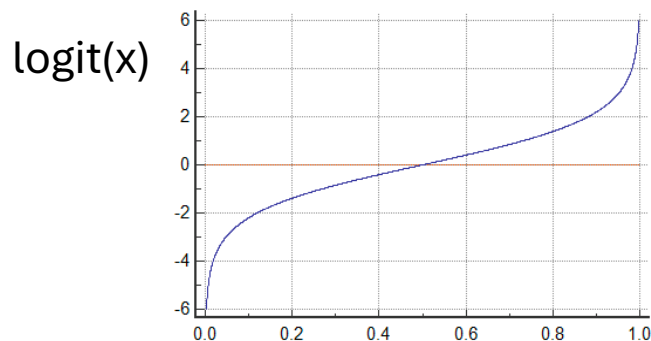


Regression example

Problem: Since all of our coefficients (β 's) can be any real numbers, our outcome $\pi(x)$ can take on any real number value! BUT our outcome of interest is the **probability**, which should only take on values between 0 and 1.

Is there something reversible we can do to $\pi(x)$ equation to move from probabilities to outputting something that ranges from $(-\infty, +\infty)$?

Solution: We can add a **link function** to map our non-linear relationship onto a linear one. This allows us to fit a linear model to our data.



$$\pi(x) = \beta_0 + \beta_1 \text{age} + \beta_2 \text{movie} + \beta_3(\text{AO3}) + \beta_4 \text{pages}$$



apply a logit transform!

$$\mathbf{logit}(\pi(x)) = \beta_0 + \beta_1 \text{age} + \beta_2 \text{movie} + \beta_3(\text{AO3}) + \beta_4 \text{pages}$$

Regression example

To save time, we re-write our model as: $\text{logit}(\pi(x)) = \beta x$

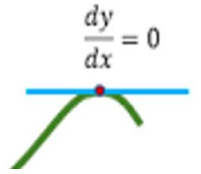
What kind of objects are β and x ?

$$\begin{pmatrix} 1 & x_{1,1} & x_{1,2} & \dots & x_{1,p} \\ 1 & x_{2,1} & x_{2,2} & \dots & x_{2,p} \\ \dots & \dots & \dots & \ddots & \dots \\ 1 & x_{n,1} & x_{n,2} & \dots & x_{n,p} \end{pmatrix}$$



ID	age	movie	Ao3	pages
1	28	Yes (1)	No (0)	12
2	59	Yes (1)	Yes (1)	0
3	30	No (0)	Yes (1)	1
4...	71	Yes (1)	Yes (1)	248

Regression example



We use our favorite statistical software to fit our model. We decide to do this by finding the maximum point of the log-likelihood function using Maximum Likelihood Estimation (MLE).

How does the software find the maximum of the log-likelihood function?

★ Differential calculus! ★

Regression example

We fit our model and get the following output.

What if we want **odds**?
Exp function!

What if we want **probabilities**?
Expit function!

	Log odds	Std. Error	Pr (> z)	Sig.
intercept	-2.91788	0.47218	6.42e-10	***
age	-0.027539	0.03420	0.320710	
movie	0.71676	0.24701	0.003712	**
ao3	1.11010	0.29775	0.000193	***
pages	0.10522	0.04134	0.010914	*

	Odds
intercept	0.05405
age	0.97284
movie	2.047787
ao3	3.03662
pages	1.11095

	Probability
intercept	0.05128
age	0.49312
movie	0.67189
ao3	0.75215
pages	0.52626

Regression example

You don't have to know all this yet!

This is just a quick introduction to some mathematical ingredients used in statistics soup





Questions?

Wrap-up



← PHS students
learning cool
content

(if they were Eridians on
40 Eridiani A b).

Resources

PHS 2000A Teaching Team

- **TFs:** Tabor Hoatson, Tillie Slosser, Ashlynn Wimer, Dylan Gould
- **Instructors:** Jarvis Chen

The PHS Office

- Eric, Jarvis, Bruce, Matthew, Claudette, Paul

Your classmates!

Resources

PHS2000A Tutoring Resources

- Available to everyone
- Email Matthew and cc Jarvis for more information

PHS2000A Canvas Site

- Has the course resources + supplemental materials
 - PHS2026 Launch Materials to reference throughout the year
- Office Hour schedule (TF + Instructor office hours)
- Information and links for academic + support resources available to you while you are a student here at Harvard

Wrap-up

Thank you to past PHS2000A Teaching Fellows who taught this material:

- Cami Christopher
- Laura Chen
- Rienna Russo
- Emma McGee
- Erin Burman
- Kat Sadikova
- Louisa Smith
- Aaron Sarvet
- Gabe Schwartz
- Bryan Patenaude

Bonus
Content
(Wooooo!!!)